

Chapter 2.A

Atoms and the Periodic Table

Chemical Principles I

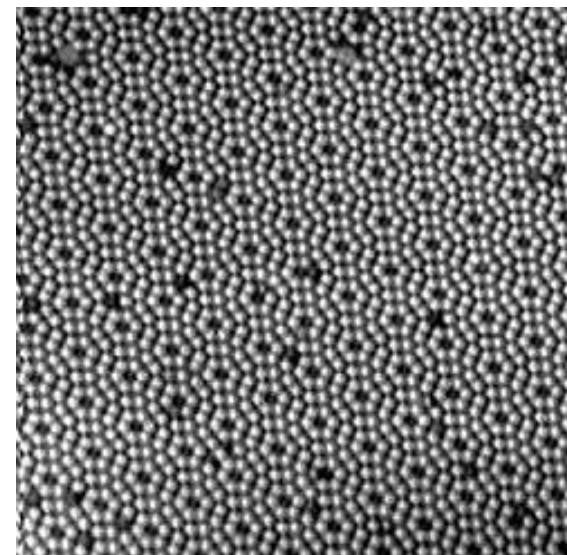
CHEM 1211K

1. Recognize and interconvert between the symbols and names of common elements.
2. Describe Dalton's atomic theory.
3. Apply Rutherford's nuclear model of the atom.
4. Identify the mass number and atomic number of an atom using the counts of subatomic particles; apply mass number and atomic number to infer counts of subatomic particles.
5. Identify isotopes of element using their atomic and mass numbers; count subatomic particles in a given isotope.
6. Calculate the atomic mass of an element from the masses and abundances of its naturally occurring isotopes.
7. Classify elements in a systematic manner based on their location on the periodic table.
8. Identify periods, groups, and sections of the periodic table.

Which statements about the differences between mixtures and compounds are *true*? Select all that apply.

- A. Compounds are pure substances and mixtures are not.
- B. Compounds can be separated into constituent components by physical means.
- C. Only mixtures can contain more than one element.
- D. The ratio of components is fixed in a compound and variable for a mixture.
- E. All compounds are mixtures. There is no difference.

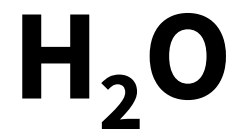
- **Atoms** are the fundamental building blocks of matter and are generally too small for us to see.
- **Scanning tunneling microscopy (STM)** and **atomic force microscopy (AFM)** can be used to image at the atomic level.
- Each of the known chemical elements is identified by a **chemical symbol** consisting of one capital letter (C, H, O, F, etc.) or one capital letter followed by a lower-case letter (Na, Fe, Cl, Ar, etc.).
- Ten elements have symbols derived from non-English names:



10 nm

English Name	Symbol	Basis	English Name	Symbol	Basis
Antimony	Sb	Stibium	Potassium	K	Kalium
Gold	Au	Aurum	Silver	Ag	Argentum
Iron	Fe	Ferrum	Sodium	Na	Natrium
Lead	Pb	Plumbum	Tin	Sn	Stannum
Mercury	Hg	Hydrargyrum	Tungsten	W	Wolfram

- We combine chemical symbols with subscripted numbers, parentheses, and other symbols to create **chemical formulas**.
- Subscript numbers in formulas indicate the number of atoms of each element present
- Parentheses are used to group atoms linked by covalent bonds (e.g., polyatomic ions)



- **The Law of Conservation of Mass:** the mass of material present at the end of a chemical reaction is equal to the mass of material present at the start of the reaction.
- **Example.** Phosphorus reacts with oxygen in the air to form an oxide of phosphorus. The total mass of oxygen, phosphorus, the phosphorus oxide, and the rest of the apparatus remains constant throughout the reaction.

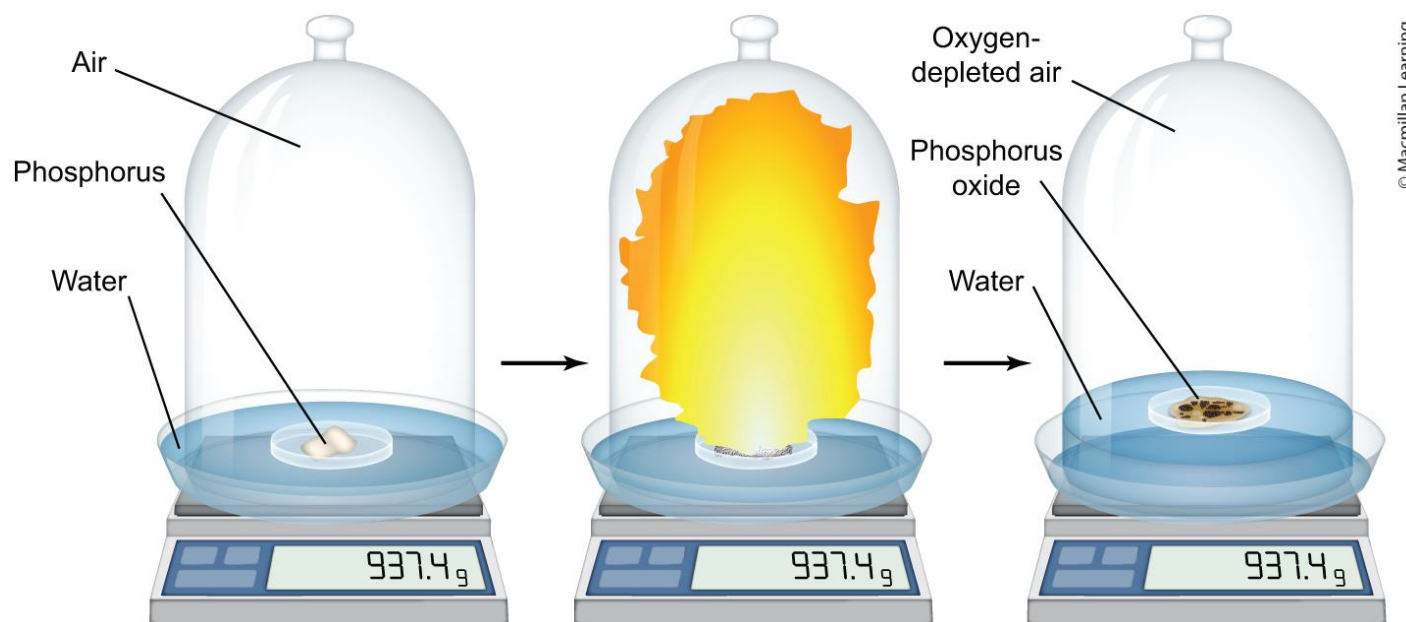


Figure 2.1

- **The Law of Definite Proportions:** every compound has a constant set of proportions by mass of the elements in that compound, which define it.
- **Example.** Every sample of pure carbon dioxide contains a 3 : 8 ratio by mass of carbon to oxygen. Any substance with carbon and oxygen in a different mass ratio cannot possibly be carbon dioxide.



- **The Law of Multiple Proportions:** different compounds may contain the same elements in different mass proportions. In that case, those mass proportions are related to each other by small, whole-number ratios.
- **Example.** Every sample of pure carbon dioxide contains a 3 : 8 ratio by mass of carbon to oxygen and every sample of carbon monoxide contains a 3 : 4 ratio by mass of carbon to oxygen. For a fixed mass of carbon in samples of these two gases, carbon dioxide contains twice the mass of oxygen as carbon monoxide.



- John **Dalton** formulated an **atomic theory** to explain the laws of chemical combination.
 1. Matter is made up of very tiny, indivisible particles called **atoms**.
 2. Each atom of a particular element has the same mass, but the mass of an atom of one element is different from the mass of an atom of any other element.
 3. Atoms combine to form what we now call **molecules**. When they do so, atoms combine in small, whole-number ratios.
 4. Atoms of some pairs of elements can combine with each other in different, small, whole-number ratios to form different compounds.

2.3 The History of the Atom | 2 of 4

- The **electron**, a negatively charged constituent of the atom (**subatomic particle**), was discovered by J. J. Thomson in the late 1800s.
- American physicist Robert Millikan succeeded in measuring the mass of a single electron in the early 1900s.
- The “non-electron” part of the atom must have a charge with positive sign equal in magnitude to the charge of the electrons, such that the atom as a whole is electrically neutral.
- In the lightest atom (hydrogen), the electron is only $1/1836^{\text{th}}$ of the total mass of the atom
- The **plum-pudding model** of the atom followed from these observations

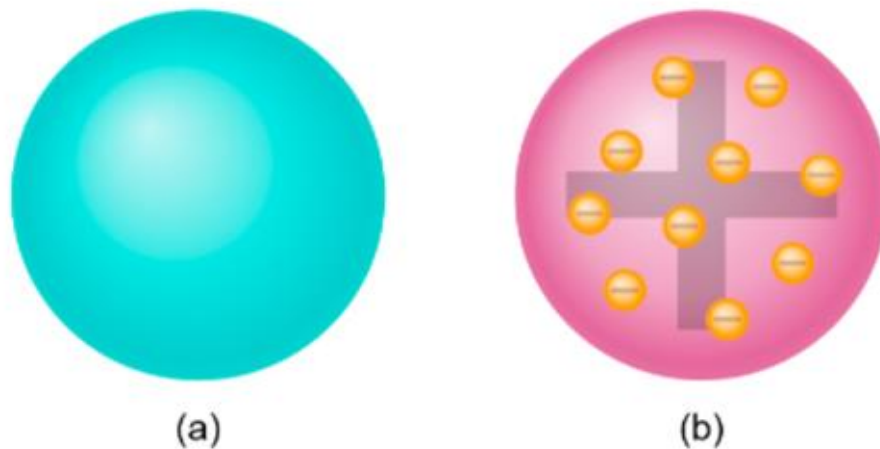


Figure 2.7. Dalton's model of the atom (a) and the plum-pudding model (b).

- By shooting alpha particles at a thin sample of gold foil, Rutherford hoped to validate the plum-pudding model
- Instead, this experiment led to a **nuclear model** for the structure of the atom, with an extremely small nucleus surrounded by a “cloud” of negatively charged electrons

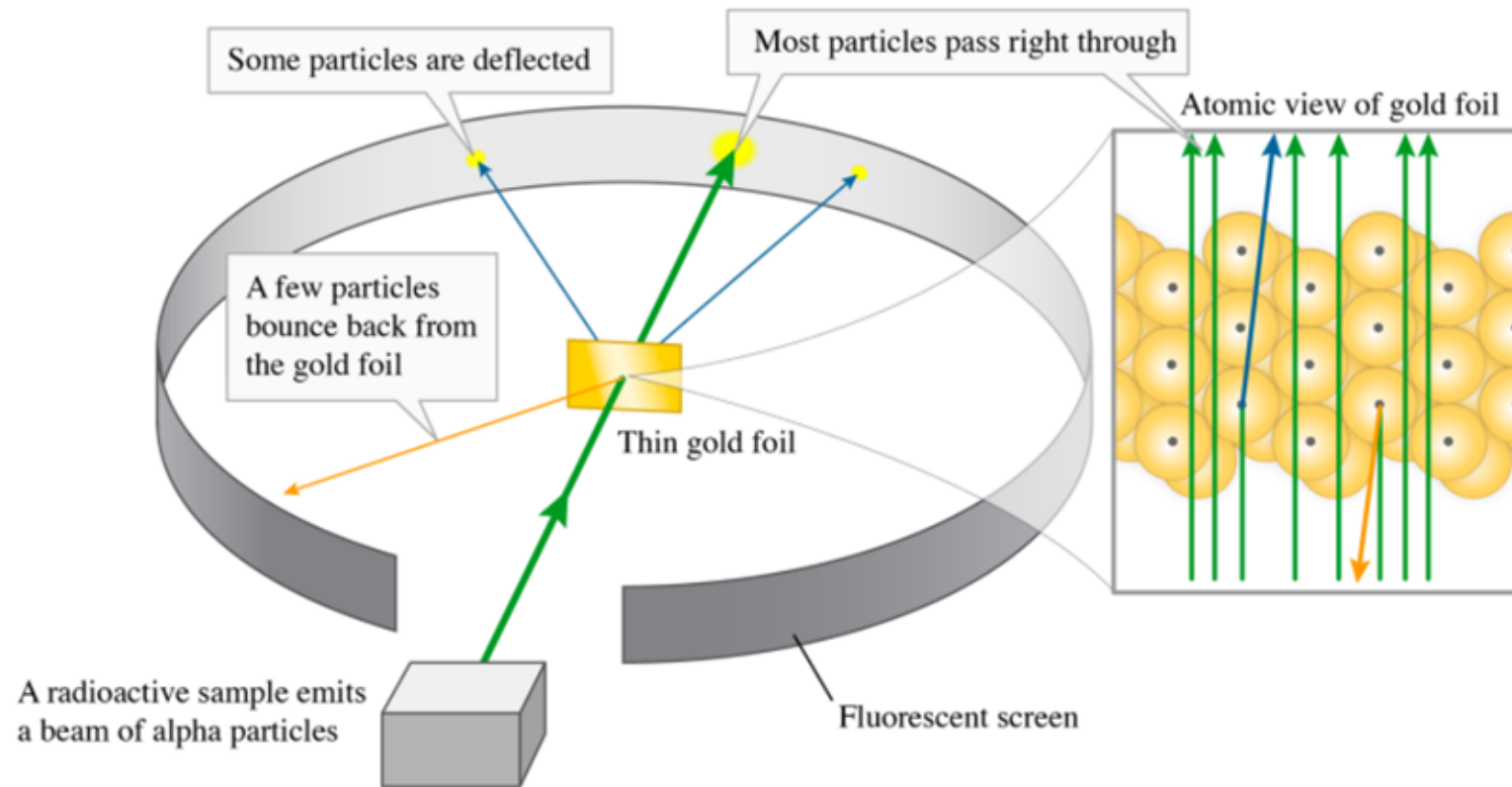


Figure 2.8. Rutherford's gold-foil experiment.

- The positive charge of the atom and most of its mass are in a relatively small area, the **nucleus**.
- Most of the atom is empty space. Light, negatively charged **electrons** are spread throughout this empty space.
- The number of negatively charged electrons is equal to the number of positively charged particles (**protons**) inside the nucleus. Neutral nuclear particles called **neutrons** account for the remaining mass of the atom.

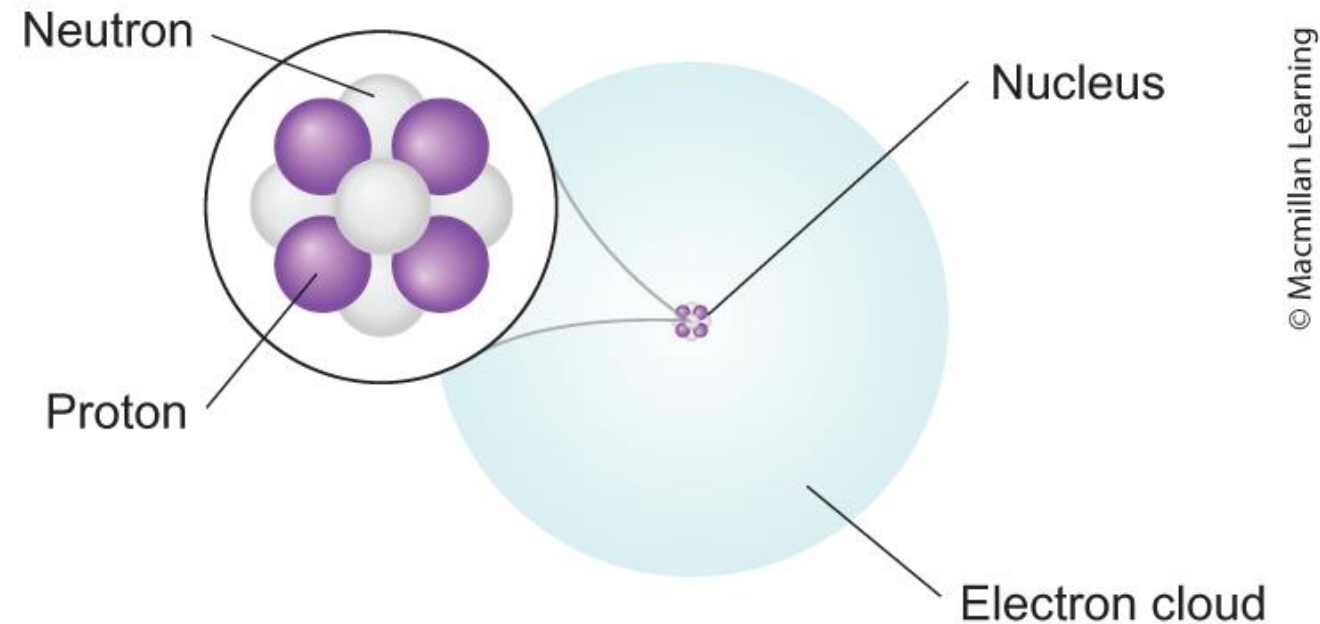


Figure 2.9. Nuclear model of the atom.

- Protons, neutrons, and electrons are the constituent particles of atoms and are known as **subatomic particles**.

Particle	Charge (e)	Mass (u)	Location
Proton	+1	1.0073	In the nucleus
Neutron	0	1.0073	In the nucleus
Electron	-1	5.49×10^{-4}	Outside the nucleus

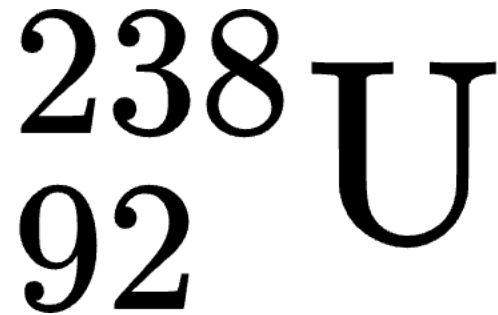
The elementary charge e is 1.6×10^{-19} Coulombs. One atomic mass unit (u or amu) is 1.67×10^{-24} grams.

Which *non-neutral* subatomic particle is located in the nucleus of the atom?

- A. Proton
- B. Neutron
- C. Electron

- Atoms are electrically neutral:
- The **atomic number (Z)** is equal to the number of protons in the nucleus of the atom (or, equivalently, the number of electrons in the neutral atom)
- The **mass number (M)** is equal to the sum of protons and neutrons in the nucleus of the atom

- Superscript and subscript numbers are used in **atomic symbols** to show M and Z , respectively



- **Isotopes** are atoms of the same element with different numbers of neutrons in the nucleus.
- Isotopes have the same atomic number (Z) but different mass numbers (M).
- **Example.** Carbon-14 and carbon-12 are isotopes.

Of the choices below, which element has the greatest atomic number?

- A. Carbon
- B. Helium
- C. Iron
- D. Sodium
- E. Bromine

How many neutrons are in the nucleus of an atom of ^{238}U ? Enter your response as a numeral.

Chapter 2.B

Atoms and the Periodic Table

Chemical Principles I

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- **Ions**
- **Anions**
- **Cations**
- Ions can either be **monatomic**, containing a single atom, or **polyatomic**, consisting of multiple atoms linked by covalent bonds.

How many electrons are in the ion below? Refer to the periodic table as needed!



- The **atomic mass unit (u or amu)** is defined as $1/12^{\text{th}}$ the mass of a carbon-12 atom.
- Atomic masses are on a *relative scale*: carbon-12 is assigned a mass of exactly 12.00 u and all other atomic masses are determined relative to carbon-12.
- We will use the term **isotopic mass** to refer to the relative mass of an atom of a specific isotope.
- Isotopic masses are not whole numbers due to the **binding energy** of the nucleus. Some portion of the total mass of all nucleons “goes into” the nuclear binding energy.

- The **average atomic mass (average atomic weight)** for an element is a weighted average of all the isotopic masses for that element, with the weights equal to the **abundances** of the isotopes.
- Can be found on the periodic table

Antimony has two stable isotopes with the following atomic masses (in atomic mass units):

$$^{123}\text{Sb} = 122.904 \text{ u}$$

$$^{121}\text{Sb} = 120.868 \text{ u}$$

Determine the isotopic abundance of ^{123}Sb using the average atomic mass of Sb (121.76 u).
Enter your response as a percentage to the nearest 0.1%.

There are three naturally occurring isotopes of silicon (atomic masses listed below).
The percent abundance of ^{29}Si is 4.68%.

Find the percent abundance of ^{30}Si given the following information and the average atomic mass of Si (28.085 u).

$$^{28}\text{Si} = 27.977 \text{ u}$$

$$^{29}\text{Si} = 28.977 \text{ u}$$

$$^{30}\text{Si} = 29.974 \text{ u}$$

- The **periodic table** is an organization of elements by increasing atomic number with elements having similar chemical properties aligned in vertical columns.
- Columns of the table are called **groups**;
the groups are numbered 1 – 18 from left to right.
- Rows of the table are called **periods**;
the periods are numbered starting with 1 at the top.

ОПЫТЪ СИСТЕМЫ ЭЛЕМЕНТОВЪ.
ОСНОВАННОЙ НА ИХЪ АТОМНОМЪ ВѢСѢ И ХИМИЧЕСКОМЪ СХОДСТВѢ.

			Ti = 50	Zr = 90	? = 180.
			V = 51	Nb = 94	Ta = 182.
			Cr = 52	Mo = 96	W = 186.
			Mn = 55	Rh = 104,4	Pt = 197,1.
			Fe = 56	Ru = 104,4	Ir = 198.
			Ni = 59	Pd = 106,6	Os = 199.
			Cu = 63,4	Ag = 108	Hg = 200.
H = 1	Be = 9,1	Mg = 24	Zn = 65,2	Cd = 112	
	B = 11	Al = 27,1	? = 68	Ur = 116	Au = 197?
	C = 12	Si = 28	? = 70	Sn = 118	
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?
	O = 16	S = 32	Se = 79,4	Te = 128?	
	F = 19	Cl = 35,5	Br = 80	I = 127	
Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204.
		Ca = 40	Sr = 87,6	Ba = 137	Pb = 207.
		? = 45	Ce = 92		
		?Er = 56	La = 94		
		?Yt = 60	Di = 95		
		?In = 75,6	Th = 118?		

Д. Менделѣевъ

Figure 2.15. The periodic table of Mendeleev.

2.6 The Periodic Table | 2 of 4

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026	
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122																	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305																	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798	
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29	
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71																86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103																118 Og Oganesson (294)
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																			
6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97				
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)				

Ptable

Figure. The modern periodic table.

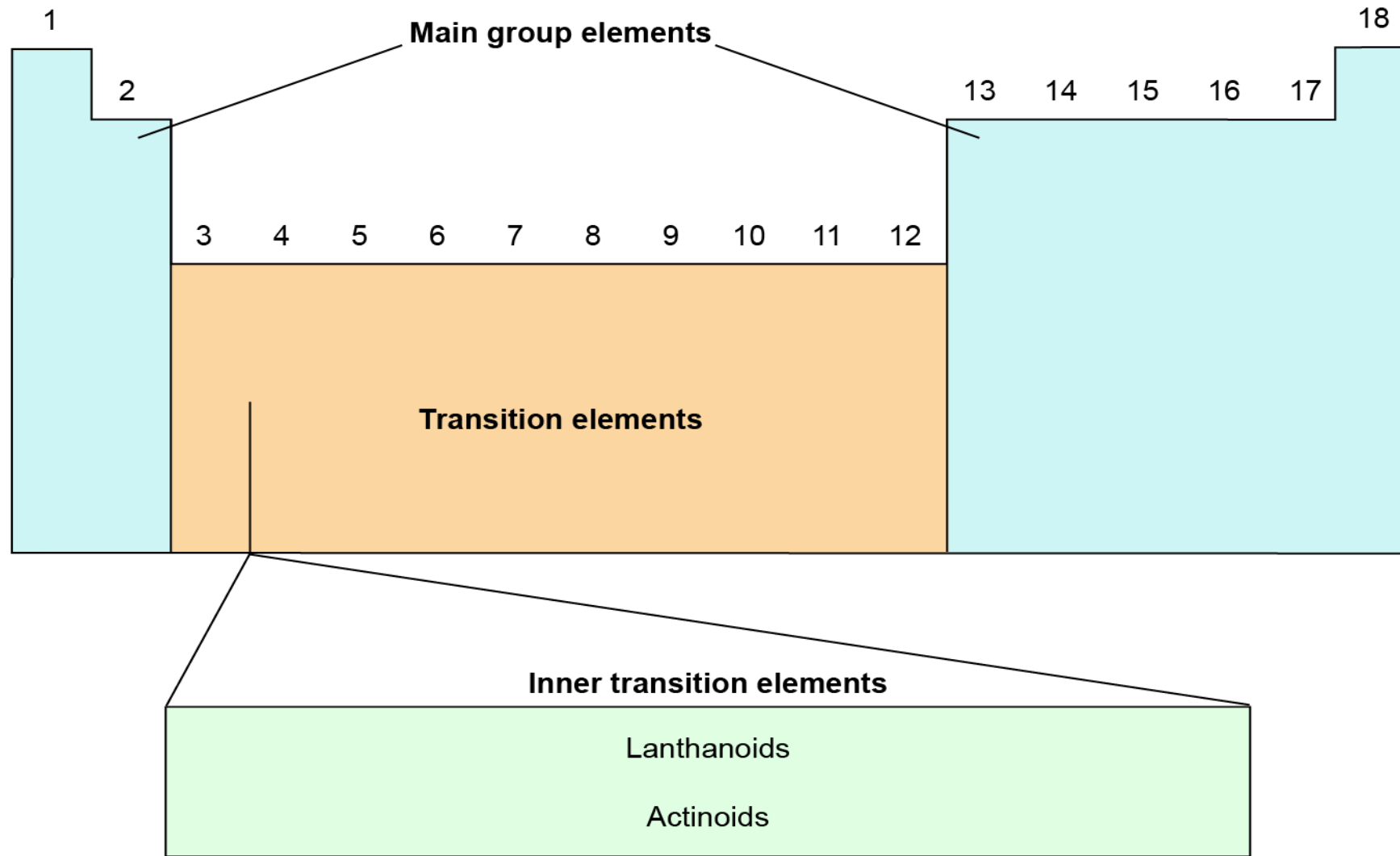


Figure 2.18. Grouping the elements into **main group** and **transition elements**.

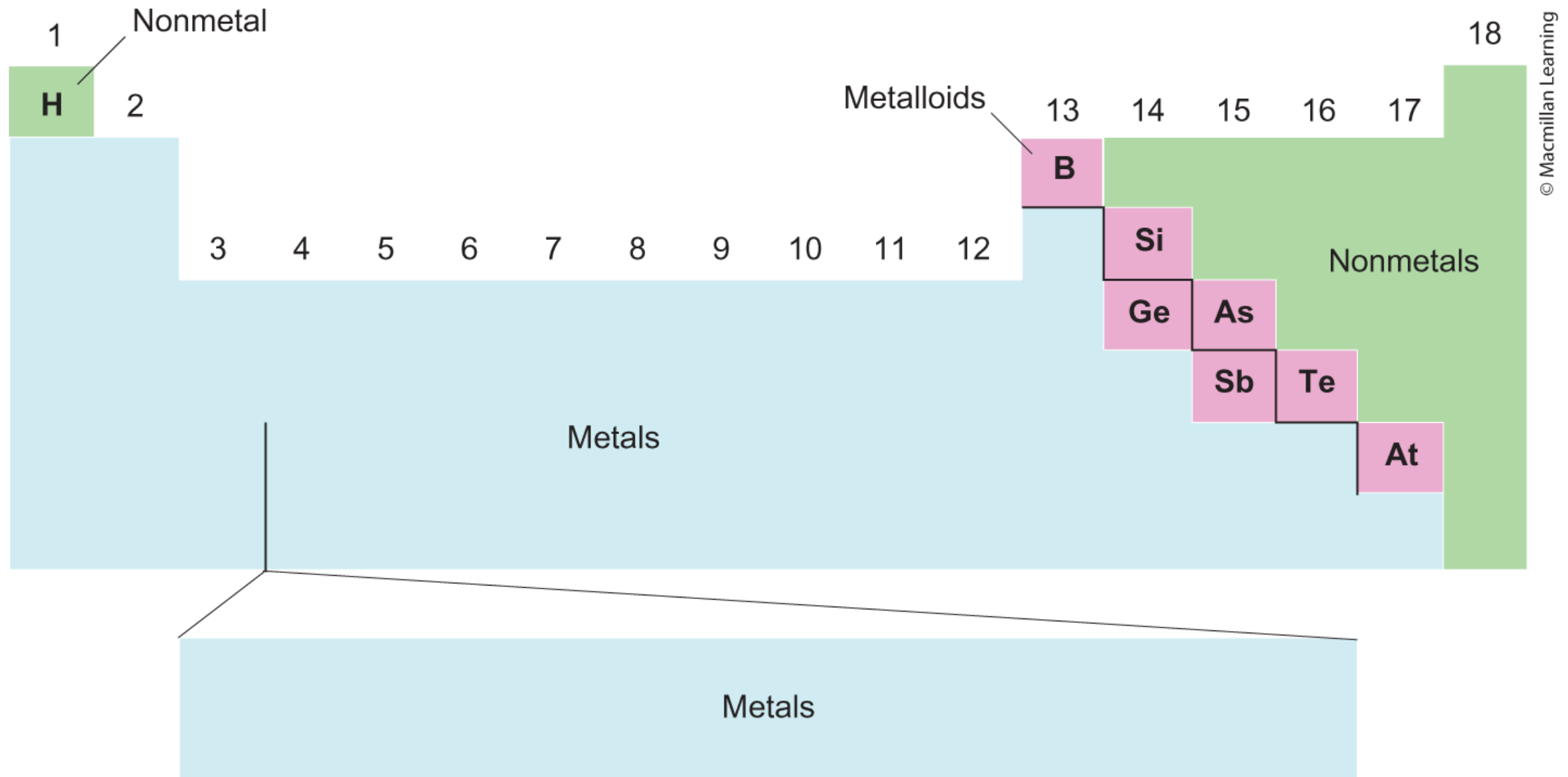


Figure 2.18. Grouping the elements into nonmetals, metals, and metalloids.

Which of the following elements are classified as nonmetals?

A. Si

B. Ne

C. Fe

D. Sr

E. Br

- Some of the amazing things we can now do with atoms and the periodic table...
 1. Represent atoms and molecules using chemical formulas.
 2. Interpret chemical formulas to count atoms.
 3. Apply the laws of chemical combination to define and distinguish between compounds and establish fundamental constraints on the nature of matter.
 4. Visualize and count the subatomic particles (protons, neutrons, and electrons) found within neutral atoms and ions.
 5. Recognize isotopes and apply isotopic masses and abundances to carry out calculations involving average atomic mass.
 6. Describe, classify, and relate elements based on their positions on the periodic table.